

Customer Data Analysis Report

Data Analysis Internship Project

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Report Type: Comprehensive Shopping Behavior Analysis

Dataset: Multi-Mall Customer Transactions

Tools Used: SQL · Power BI · Excel

Year: 2025–2026

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Executive Summary

This report presents a comprehensive analysis of customer shopping behavior across multiple shopping malls. The dataset captures 99,000+ transactions totalling **251.51 million TL** in revenue, covering 299,000 products sold across eight product categories. Using SQL queries and Power BI dashboards, the analysis explores shopping patterns segmented by **gender**, **age group**, **product category**, and **payment method**.



KEY SUMMARY FINDINGS

- **Female customers** account for 59.81% of transactions and ~60% of revenue.
- **Clothing** is the top-selling category across all genders and age groups.
- **55+ age group** is the most active segment by transactions, products, and revenue.
- **Cash** is the most preferred payment method at 44.69% of transactions.
- Digital payments (Credit + Debit Card) together exceed 55% of all transactions.

Dashboard Overview

The following Power BI dashboards provide an interactive visual summary of the customer shopping dataset. Three dashboard views are presented: the main overview, gender & age analysis, and payment & category analysis.

Dashboard 1 — Customer Shopping Overview

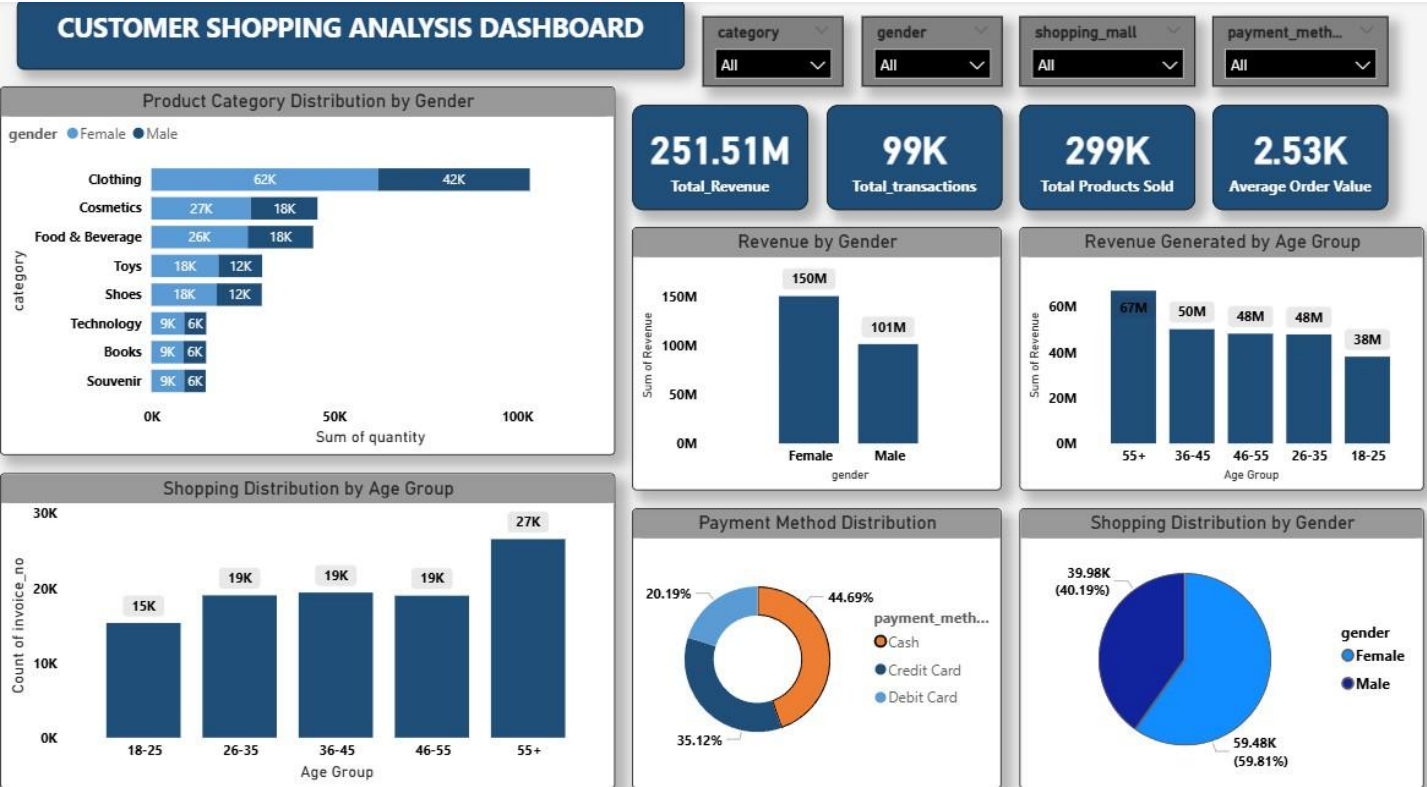


Figure 1: Main dashboard showing KPIs, product category distribution, revenue by gender/age, payment method distribution, and shopping activity by age group.

Dashboard 2 — Gender & Age Analysis



Figure 2: Gender & Age dashboard showing products sold, revenue, and category distribution broken down by gender and age group segments.

Dashboard 3 — Payment & Category Analysis

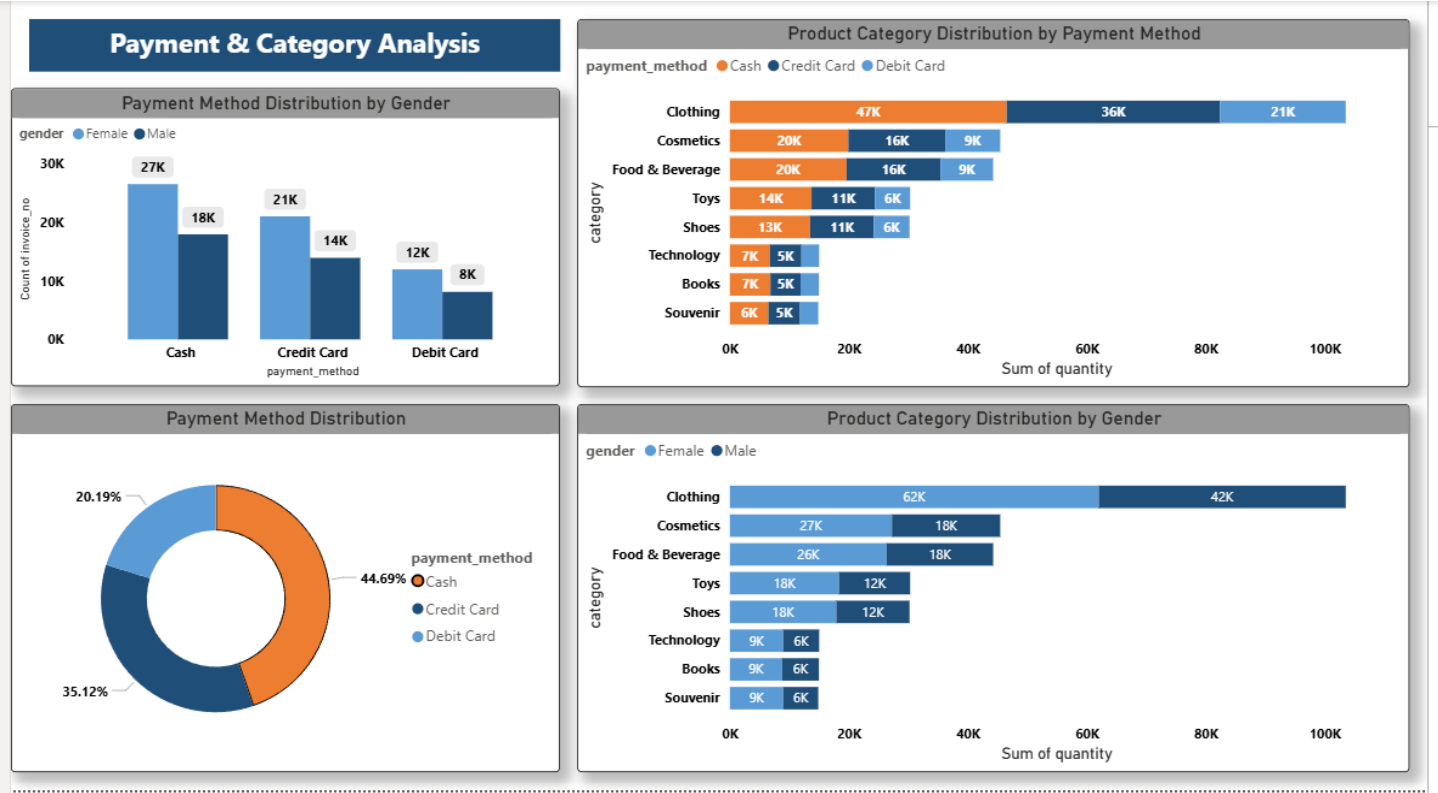


Figure 3: Payment & Category dashboard showing payment method distribution by gender and product category across all payment types.

Q1 How is the shopping distribution according to gender?

OBJECTIVE
Understand the proportion of male and female customers in the dataset and identify the dominant customer segment.

SQL Query

```
SELECT gender,
COUNT(*) AS total_customers
FROM customer
GROUP BY gender;
```

Query Output

Gender	Total Transactions
Female	59,482
Male	39,975

Analysis & Insight

Female customers account for **59.81%** of total transactions while male customers contribute **40.19%**. This indicates that female shoppers represent the dominant customer segment across all shopping malls in the dataset. The higher transaction count suggests female customers engage more frequently in shopping activities.

KEY FINDING
Female customers generated 19,507 more transactions than male customers, making them the primary shopping demographic.

Recommendations

- Design targeted marketing campaigns for female customers.
- Increase inventory availability for products frequently purchased by female shoppers.
- Introduce loyalty and reward programs to encourage repeat purchases.
- Analyze product category preferences among female customers for better merchandising.

Q2 Which gender did we sell more products to?

OBJECTIVE
Identify which gender purchased the highest quantity of products and understand customer buying behavior.

SQL Query

```
SELECT gender,
SUM(quantity) AS total_products
FROM customer
GROUP BY gender;
```

Query Output

Gender	Total Products Sold
Female	178,659
Male	120,053

Analysis & Insight

Female customers purchased **178,659 products** compared to **120,053** by male customers. Female customers accounted for approximately 59.8% of total products sold, confirming they not only transact more but also buy higher volumes per visit.

KEY FINDING
Female customers purchased 58,606 more products than male customers — making them the largest product-consuming segment.

Recommendations

- Focus promotional campaigns on female customer segments.
- Maintain sufficient inventory for products frequently purchased by female shoppers.
- Analyze top-selling categories among female customers to identify growth opportunities.
- Launch loyalty programs and personalized offers to increase repeat purchases.

Q3 Which gender generated more revenue?

OBJECTIVE
Identify which gender contributes the highest revenue and understand the revenue-generating customer segment.

SQL Query

```
SELECT gender,
SUM(price * quantity) AS revenue
FROM customer
GROUP BY gender
ORDER BY revenue;
```

Query Output

Gender	Revenue (TL)
Male	101,298,658.23
Female	150,207,136.02

Analysis & Insight

Female customers generated approximately **150.21 million TL**, while male customers generated approximately **101.30 million TL**. Female customers contributed nearly **60%** of total revenue, making them the highest revenue-generating segment. This aligns with findings from Q1 and Q2 — female shoppers transact more frequently and purchase higher volumes, naturally translating into greater revenue.

KEY FINDING
Female customers generated approximately 48.91 million TL more revenue than male customers.

Recommendations

- Prioritize marketing campaigns targeted toward female customers.
- Develop personalized offers and loyalty programs for high-value female shoppers.
- Ensure adequate inventory for categories generating high female revenue.
- Analyze premium product preferences for upselling and cross-selling opportunities.

Q4 Distribution of purchase categories relative to gender?

OBJECTIVE

Analyze how product category purchases are distributed across male and female customers and identify the most popular categories for each gender.

SQL Query

```
SELECT gender, category,
SUM(quantity) AS products_sold
FROM customer
GROUP BY gender, category;
```

Query Output

Gender	Category	Products Sold
Female	Clothing	62,039
Female	Cosmetics	27,261
Female	Food & Beverage	26,362
Female	Toys	18,362
Female	Shoes	17,906
Male	Clothing	41,519
Male	Cosmetics	18,204
Male	Food & Beverage	17,915

Analysis & Insight

Clothing is the most purchased category for both genders. Female customers purchased 62,039 clothing products vs 41,519 by male customers. Female customers consistently outpurchase males across every product category. Technology, Books, and Souvenir categories recorded the lowest volumes for both genders.

KEY FINDINGS

- Clothing is the top-selling category for both genders.
- Female customers dominate purchases across every category.
- Cosmetics and Food & Beverage are the next strongest categories.
- Technology, Books, and Souvenirs have the lowest demand.

Recommendations

- Maintain higher inventory for Clothing due to consistently strong demand.
- Develop category-specific promotions aimed at female customers, especially in Clothing and Cosmetics.

- Increase promotional efforts for lower-performing categories: Books, Technology, and Souvenirs.
- Bundle Clothing with Cosmetics and Accessories to increase average transaction value.

Q5 How is the shopping distribution according to age?

OBJECTIVE
Analyze shopping activity across different age groups and identify which age segment contributes the highest number of transactions.

SQL Query

```
SELECT
CASE
WHEN age BETWEEN 18 AND 25 THEN '18-25'
WHEN age BETWEEN 26 AND 35 THEN '26-35'
WHEN age BETWEEN 36 AND 45 THEN '36-45'
WHEN age BETWEEN 46 AND 55 THEN '46-55'
ELSE '55+'
END AS age_group,
COUNT(*) AS transactions
FROM customer
GROUP BY age_group;
```

Query Output

Age Group	Transactions
18-25	15,359
26-35	19,059
36-45	19,436
46-55	19,016
55+	26,587

Analysis & Insight

The **55+ age group** recorded the highest transactions with **26,587 purchases**. The 26-35, 36-45, and 46-55 groups each contribute ~19,000 transactions, showing stable middle-age engagement. The 18-25 group recorded the fewest transactions at 15,359, suggesting lower youth engagement with physical mall shopping.

KEY FINDING
The 55+ segment dominates shopping activity with 73% more transactions than the youngest group (18-25).

Recommendations

- Develop loyalty programs and personalized offers for customers aged 55+.
- Continue targeted promotions for the 26-55 age segments.

- Introduce digital marketing campaigns and student offers to attract 18–25 customers.
- Create age-specific product recommendations and campaigns based on customer preferences.

Q6 Which age category did we sell more products to?

OBJECTIVE
Identify which age group purchased the highest quantity of products and understand product consumption patterns across age segments.

SQL Query

```
SELECT
CASE
WHEN age BETWEEN 18 AND 25 THEN '18-25'
WHEN age BETWEEN 26 AND 35 THEN '26-35'
WHEN age BETWEEN 36 AND 45 THEN '36-45'
WHEN age BETWEEN 46 AND 55 THEN '46-55'
ELSE '55+'
END AS age_group,
SUM(quantity) AS products
FROM customer
GROUP BY age_group;
```

Query Output

Age Group	Products Sold
18-25	46,028
26-35	57,265
36-45	58,483
46-55	57,086
55+	79,850

Analysis & Insight

The **55+ age group** purchased the highest number of products at **79,850 units**. Middle-aged customers (26-55) each purchased between 57,000-58,000 products, showing strong stable demand. The 18-25 group purchased the fewest at 46,028. Product purchases positively correlate with age.

KEY FINDING
The 55+ group purchased 73.5% more products than the 18-25 group — 79,850 vs 46,028 units.

Recommendations

- Introduce exclusive loyalty programs for customers aged 55 and above.
- Optimize inventory for products frequently purchased by the 55+ segment.

- Design personalized campaigns based on age-specific category preferences.
- Create promotional offers and student discounts to boost 18–25 purchasing activity.

Q7 Which age category generated more revenue?

OBJECTIVE

Identify the age group contributing the highest revenue and understand which customer segment has the greatest impact on profitability.

SQL Query

```
SELECT
CASE
WHEN age BETWEEN 18 AND 25 THEN '18-25'
WHEN age BETWEEN 26 AND 35 THEN '26-35'
WHEN age BETWEEN 36 AND 45 THEN '36-45'
WHEN age BETWEEN 46 AND 55 THEN '46-55'
ELSE '55+'
END AS age_group,
SUM(quantity * price) AS revenue
FROM customer
GROUP BY age_group;
```

Query Output

Age Group	Revenue (TL)
18-25	38,118,271.35
26-35	47,879,659.58
36-45	50,184,235.79
46-55	48,219,742.64
55+	67,103,884.89

Analysis & Insight

The **55+ age group** generated the highest revenue at approximately **67.10 million TL**. The 36-45 group is the second most valuable segment with 50.18 million TL. The 18-25 group contributed the least at 38.12 million TL, reflecting both fewer transactions and lower per-transaction spend. Revenue increases positively with customer age.

KEY FINDING

55+ customers generated 75.9% more revenue than 18-25 customers (67.10M vs 38.12M TL).

Recommendations

- Develop premium loyalty programs for 55+ customers.
- Target the 36-45 and 46-55 segments with upselling and category-specific promotions.

- Introduce affordable bundles and digital campaigns to increase 18–25 spending.
- Create personalized membership programs to retain high-value senior customers.

Q8

Distribution of purchase categories relative to age group?

OBJECTIVE
Analyze how product category preferences vary across age groups and identify the most popular categories within each customer segment.

SQL Query

```
SELECT
CASE
WHEN age BETWEEN 18 AND 25 THEN '18-25'
...
END AS age_group,
category,
SUM(quantity) AS products_sold
FROM customer
GROUP BY age_group, category;
```

Key Query Results

Age Group	Top Category	Products Sold
18–25	Clothing	15,851
26–35	Clothing	19,820
36–45	Clothing	20,054
46–55	Clothing	~20K
55+	Clothing	~28K

Analysis & Insight

Clothing is the most purchased category across every age group. The 55+ segment drives the highest category sales in nearly all product types, particularly Clothing, Cosmetics, and Food & Beverage. Technology, Books, and Souvenirs consistently have the lowest demand regardless of age.

- KEY FINDINGS**
- Clothing leads all categories across all age groups.
 - 55+ customers are the top buyers in almost every category.
 - Cosmetics and Food & Beverage are strong secondary categories across all ages.
 - Technology, Books, and Souvenirs underperform across every segment.

Recommendations

- Prioritize Clothing stock availability due to universally strong demand.

- Design targeted promotions for 55+ customers who lead in category sales.
- Introduce complementary bundles within Clothing and Cosmetics to increase basket size.
- Run seasonal campaigns to promote Books, Technology, and Souvenir products.

Q9 Does the payment method have a relation with other columns?

OBJECTIVE
Analyze whether customer payment preferences vary across different genders and product categories and identify meaningful relationships between payment methods and purchasing behavior.

SQL Query 1 — Payment Method vs Gender

```
SELECT payment_method, gender,
COUNT(*) AS transactions
FROM customer
GROUP BY payment_method, gender;
```

Payment Method	Female	Male
Cash	26,500	17,900
Credit Card	21,000	13,900
Debit Card	12,000	8,100

SQL Query 2 — Payment Method vs Product Category

```
SELECT payment_method, category,
SUM(quantity) AS products_sold
FROM customer
GROUP BY payment_method, category;
```

Payment Method	Top Category	Products Sold
Cash	Clothing	47K
Credit Card	Clothing	36K
Debit Card	Clothing	21K

Analysis & Insight

Cash is the most frequently used payment method for both genders. Female customers record higher transaction counts across all payment methods, consistent with their overall shopping dominance. Clothing remains the top category regardless of payment type. Cash transactions dominate product sales in every category. Debit Card usage is lowest across all segments.

KEY FINDINGS

- Cash is the most preferred payment method for both genders.
- Female customers use all payment methods more frequently than males.
- Clothing leads sales across all three payment methods.
- Debit Card contributes the smallest share of category sales.

Recommendations

- Offer discounts or cashback for Credit Card and Debit Card transactions to boost digital adoption.
- Maintain robust cash-handling infrastructure since Cash remains the dominant method.
- Run category-specific promotions linked to digital payment methods.
- Introduce loyalty rewards for Credit Card and Debit Card users.

Q10 How is the distribution of the payment method?

OBJECTIVE
Analyze the overall distribution of customer payment methods and identify the most preferred mode of payment across all transactions.

SQL Query

```
SELECT payment_method,
COUNT(*) AS transactions
FROM customer
GROUP BY payment_method;
```

Query Output

Payment Method	Transactions	Share (%)
Cash	44,447	44.69%
Credit Card	34,931	35.12%
Debit Card	20,079	20.19%

Analysis & Insight

Cash is the most preferred payment method at **44.69%** of all transactions. Credit Card is second at **35.12%**, indicating strong digital payment adoption. Debit Card accounts for **20.19%**. Combined, digital payments (Credit + Debit Card) account for more than **55%** of total transactions, demonstrating a significant and growing shift toward electronic payment methods.

- KEY FINDINGS**
- Cash dominates at 44.69% — still the single most used method.
 - Credit Card usage is strong at 35.12%, nearly matching cash.
 - Digital payments combined exceed 55% of all transactions.
 - Debit Card remains the least preferred individual method at 20.19%.

Recommendations

- Offer cashback, discounts, and reward points for Credit Card and Debit Card transactions.
- Continue supporting cash payments as they remain the most used single method.
- Collaborate with banks and payment providers to increase digital payment adoption.
- Ensure fast, secure payment options across all checkout channels.

Key Insights & Strategic Recommendations

GENDER INSIGHTS	AGE INSIGHTS
- Females = 59.81% of transactions - Females generated 150.21M TL revenue - Females purchased 178,659 products - Female dominance spans all categories	55+ group leads in transactions, products & revenue 55+ generated 67.10M TL (highest of all groups) 18–25 group is consistently lowest across all metrics - Revenue positively correlates with age
CATEGORY INSIGHTS	PAYMENT INSIGHTS
- Clothing #1 across all genders & age groups - Cosmetics & Food & Beverage are strong runners-up - Technology, Books & Souvenirs underperform 55+ customers drive category demand	- Cash = 44.69% of transactions (most preferred) - Digital payments combined = 55.31% - Debit Card has the lowest individual usage - Payment preference is consistent across genders

Strategic Recommendations Summary

■ Target Female Customers

Female shoppers drive 60% of revenue. Tailor marketing, inventory, and loyalty programs to this segment.

■ Retain 55+ Customers

This age group leads in transactions, products, and revenue. Invest in senior loyalty programs and personalized offerings.

■ Boost Youth Engagement

The 18–25 group underperforms. Digital marketing, student discounts, and social media campaigns are recommended.

■ Optimize Clothing & Cosmetics Stock

These are the top-performing categories across all demographics. Avoid stock-outs.

■ Accelerate Digital Payment Adoption

Digital payments already exceed 55%. Cashback and reward programs for card users can shift the balance further.

■ Promote Low-Performing Categories

Technology, Books, and Souvenirs need targeted seasonal campaigns and bundle offers to increase visibility.